Tutorial 6: Sequence Diagrams

MCA Software Engineering

Learning Goals

Understand what a sequence diagram shows.

Identify actor, lifeline, message, activation and return message.

Understand alt (choice) and loop (repetition).

Read messages from top to bottom.

Prepare a simple sequence diagram for a software scenario.

What is a Sequence Diagram?

A sequence diagram shows how people and software objects communicate step by step.

It tells us WHO sends a message, WHO receives it, and WHEN it happens.

Time moves from top to bottom.

Example: Student → Login System → Database → Student.

Actor and Lifeline

Actor = person or external system that takes part in the scenario.

Examples: Student, Member, Librarian, Admin.

Lifeline = vertical line below an actor or object.

The lifeline shows that the participant is involved over time.

Messages

A message shows a request or communication between lifelines.

Write short and meaningful names.

Examples: validateUser(), checkBook(), createLoan(), showResult().

Messages near the top happen before messages lower in the diagram.

Activation and Return Message

Activation bar = the object is currently doing some work.

Return message = the result comes back to the caller.

A return message is often shown with a dashed arrow.

Not every simple diagram needs every return message.

Example: checkBook() → processing → Available

alt Fragment = IF–ELSE

Use alt when the system has different possible paths.

Example: [valid login] → openDashboard().

Alternative: [invalid login] → showError().

Write the condition clearly inside square brackets.

alt = choice / condition

loop Fragment = Repeat

Use loop when the same action happens again and again.

Example: issue several books.

Inside loop [for each book]: checkBook() → createLoan() → showDueDate().

This avoids drawing the same messages many times.

loop = repetition

Worked Example: Issue Book

1. Member requests a book.
2. Librarian enters member and book details.
3. System validates the member.
4. System checks book availability.
5. If available, system creates the loan.
6. System shows the due date.

If unavailable, an alternative path shows an error.

Read the sequence from top to bottom.

How to Draw a Sequence Diagram

Step 1: Choose one scenario.

Step 2: Identify actor and participating objects.

Step 3: Place lifelines from left to right.

Step 4: Add messages from top to bottom in real order.

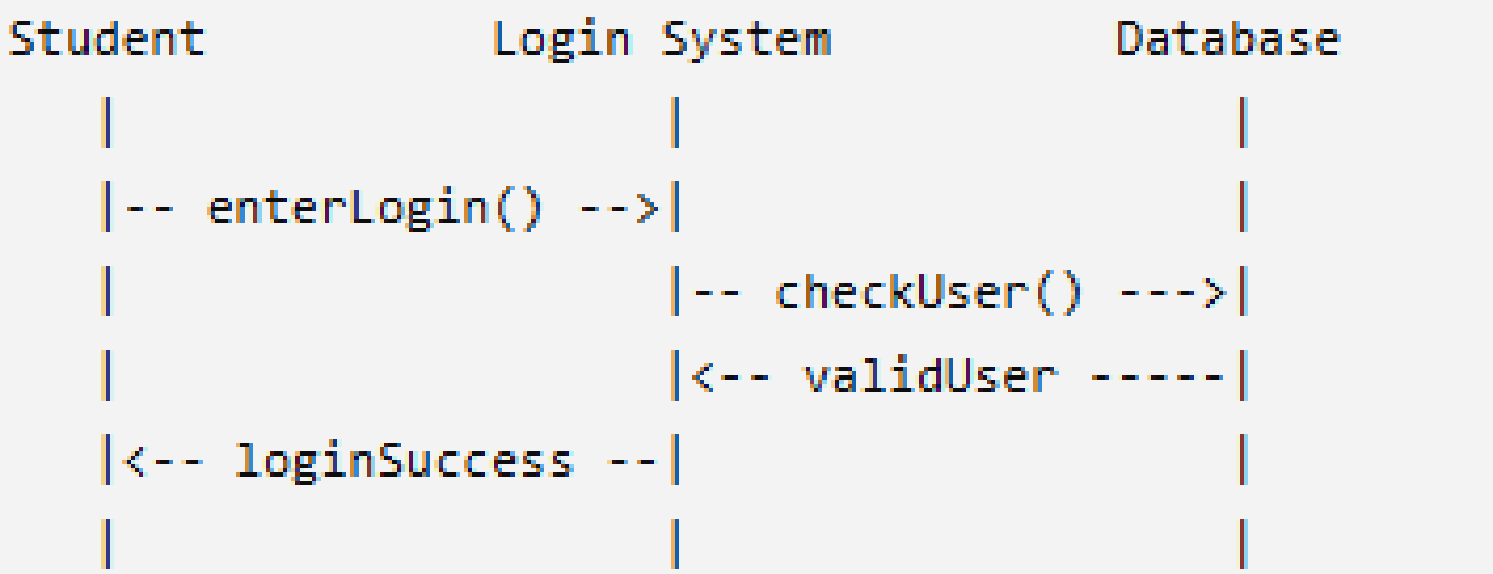
Step 5: Add alt or loop only when needed.

Step 6: Check that the diagram matches the requirements.

Example of Sequence Diagram

Suppose a student logs into a system.

Participants: Student, Login System, Database



Connection to Lab 6

Use your own project title.

Choose THREE important project scenarios.

For each scenario, identify actor(s), lifelines and messages.

Prepare THREE sequence diagrams.

Use alt/loop where needed and explain your choices during viva.